

# 实验报告：从源码构建并在干净环境安装 Wheel

| 项目   | 内容                          |
|------|-----------------------------|
| 课程名称 | 系统开发工具基础                    |
| 实验主题 | Packaging and Shipping Code |
| 题目编号 | 第9题                         |
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| 实验日期 | 2026-09-07                  |

## 一、实验目的

1. 掌握使用 `src` 布局创建最小 Python 命令行包的方法。
2. 学会使用 `python -m build` 从源码构建 wheel。
3. 掌握在干净虚拟环境中仅从 wheel 安装并验证命令行入口。

## 二、实验环境

- 操作系统：Ubuntu 25.04 (x86\_64, 内核 6.14.0-32-generic)
- 命令行工具：bash、python3.13、venv、pip、build 1.6.0
- 其他软件：setuptools (>=68)

## 三、实验步骤与代码

### 步骤1：创建 greetlab 包源码与 pyproject.toml

在 `~/lab/q09` 中创建 `src/greetlab/__init__.py`、`src/greetlab/cli.py` 和 `pyproject.toml`，把包名中的"学号"替换为 25020007191。

```
mkdir -p ~/lab/q09/src/greetlab
touch src/greetlab/__init__.py
```

src/greetlab/cli.py :

```
import argparse
def main():
    p = argparse.ArgumentParser()
    p.add_argument("--name", required=True)
    a = p.parse_args()
    print(f"Hello, {a.name}!")
```

pyproject.toml :

```
[build-system]
requires = ["setuptools ≥ 68"]
build-backend = "setuptools.build_meta"
[project]
name = "greetlab-25020007191"
version = "0.1.0"
[project.scripts]
sdt-greet = "greetlab.cli:main"
```

## 步骤2：使用 build 工具构建 wheel

```
cd ~/lab/q09 && ~/lab/venv/bin/python -m build
```

输出结果：

```
running install
running install_lib
creating build/bdist.linux-x86_64/wheel
...
Successfully built greetlab_25020007191-0.1.0.tar.gz and greetlab_25020007191-
0.1.0-py3-none-any.whl
```

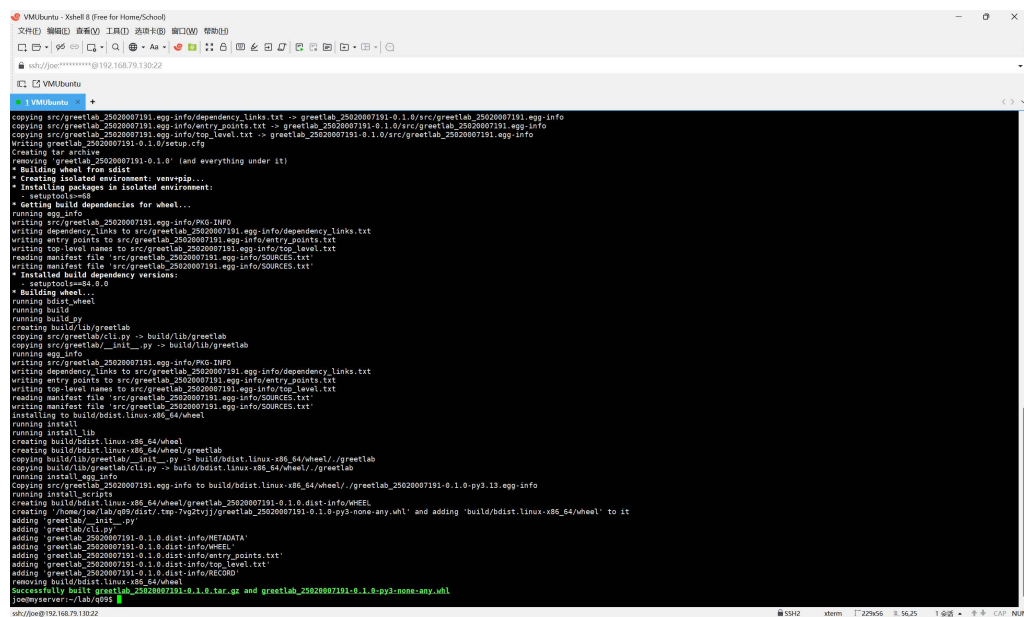
```
ls -la dist/
```

输出结果：

```
total 16
-rw-rw-r-- 1 joe joe 1776 Sep  7 05:44 greetlab_25020007191-0.1.0-py3-none-any.whl
-rw-rw-r-- 1 joe joe 1090 Sep  7 05:44 greetlab_25020007191-0.1.0.tar.gz
```

### 步骤3：新建干净虚拟环境，只从生成的 wheel 安装

```
cd ~
python3 -m venv ~/lab/q09_cleanenv
~/lab/q09_cleanenv/bin/pip install ~/lab/q09/dist/greetlab_25020007191-0.1.0-py3-
none-any.whl
~/lab/q09_cleanenv/bin/pip list
```



```
VMUbuntu - Xshell 8 (Free for Home/School)
文件(F) 编辑(E) 查看(V) 工具(T) 连接(C) 窗口(W) 帮助(H)
ssh://joe@192.168.79.13022
1 VMUbuntu
copying src/greetlab_25020007191.egg-info/dependency_links.txt --> greetlab_25020007191-0.1.0/src/greetlab_25020007191.egg-info
copying src/greetlab_25020007191.egg-info/entry_points.txt --> greetlab_25020007191-0.1.0/src/greetlab_25020007191.egg-info
copying src/greetlab_25020007191.egg-info/top_level.txt --> greetlab_25020007191-0.1.0/src/greetlab_25020007191.egg-info
Writing greetlab_25020007191-0.1.0/setup.cfg
Creating src archive
removing 'greetlab_25020007191-0.1.0' (and everything under it)
* Building wheel from sdist
* Creating isolated environment: venv+pip...
* Installing packages in isolated environment:
  setup tools==4.0
* Getting build dependencies for wheel...
running egg_info
writing src/greetlab_25020007191.egg-info/PKG-INFO
writing dependency_links to src/greetlab_25020007191.egg-info/dependency_links.txt
writing entry points to src/greetlab_25020007191.egg-info/entry_points.txt
writing top-level names to src/greetlab_25020007191.egg-info/top_level.txt
reading manifest file 'src/greetlab_25020007191.egg-info/SOURCES.txt'
writing manifest file 'src/greetlab_25020007191.egg-info/SOURCES.txt'
* Installing build dependency versions:
  setup tools==4.0
* Building wheel...
running build
running build_wheel
running build_py
creating build/lib/greetlab
copying src/greetlab/cli.py --> build/lib/greetlab
copying src/greetlab/__init__.py --> build/lib/greetlab
running egg_info
writing src/greetlab_25020007191.egg-info/PKG-INFO
writing dependency_links to src/greetlab_25020007191.egg-info/dependency_links.txt
writing entry points to src/greetlab_25020007191.egg-info/entry_points.txt
writing top-level names to src/greetlab_25020007191.egg-info/top_level.txt
reading manifest file 'src/greetlab_25020007191.egg-info/SOURCES.txt'
writing manifest file 'src/greetlab_25020007191.egg-info/SOURCES.txt'
installing to build/bdist.linux.x86_64/wheel
running install
running install_lib
creating build/bdist.linux.x86_64/wheel
creating build/bdist.linux.x86_64/wheel/greetlab
copying build/lib/greetlab/__init__.py --> build/bdist.linux.x86_64/wheel/greetlab
copying build/lib/greetlab/cli.py --> build/bdist.linux.x86_64/wheel/greetlab
running install_egg_info
Copying src/greetlab_25020007191.egg-info to build/bdist.linux.x86_64/wheel/./greetlab_25020007191-0.1.0-py3.13.egg-info
running install_scripts
creating build/bdist.linux.x86_64/wheel/greetlab_25020007191-0.1.0.dist-info/METADATA
creating /home/joe/lab/q09/dist/tmp7gztvj/greetlab_25020007191-0.1.0-py3-none-any.whl and adding 'build/bdist.linux.x86_64/wheel' to it
adding 'greetlab/__init__.py'
adding 'greetlab/cli.py'
adding 'greetlab_25020007191-0.1.0.dist-info/METADATA'
adding 'greetlab_25020007191-0.1.0.dist-info/METADATA'
adding 'greetlab_25020007191-0.1.0.dist-info/entry_points.txt'
adding 'greetlab_25020007191-0.1.0.dist-info/top_level.txt'
adding 'greetlab_25020007191-0.1.0.dist-info/RECORD'
removing build/bdist.linux.x86_64/wheel
Successfully built greetlab_25020007191-0.1.0.tar.gz and greetlab_25020007191-0.1.0-py3-none-any.whl
joe@ubuntu:~/lab/q09$
```

### 输出结果：

| Package              | Version |
|----------------------|---------|
| greetlab-25020007191 | 0.1.0   |
| pip                  | 25.0    |

### 步骤4：切到 q09 之外运行 sdt-greet

```
cd /tmp && ~/lab/q09_cleanenv/bin/sdt-greet --name 25020007191
```

### 输出结果：

```
Hello, 25020007191!
```

## 四、实验结果

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### 4.1 结果展示

wheel 成功生成并安装到干净 venv ( `pip list` 中仅有 greetlab 与 pip 自身) , 命令行入口 `sdt-greet` 在 q09 目录之外运行正常, 输出 `Hello, 25020007191!` , 退出码为 0。

### 4.2 结果分析

- `python -m build` 同时产出 sdist (.tar.gz) 和 wheel (.whl) , wheel 为 `py3-none-any` (纯 Python、平台无关) 。
- 干净 venv 中仅安装 wheel 一个包, 说明 wheel 自包含, 无额外运行时依赖。
- `[project.scripts]` 生成的可执行入口 `sdt-greet` 通过 `console_scripts` 机制指向 `greetlab.cli:main` , 验证了安装正确。

## 五、实验总结

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通过本次实验, 我掌握了以下技能:

1. 使用 `src` 布局与 `pyproject.toml` 声明构建系统、项目元数据和脚本入口。
2. 使用 `python -m build` 构建与使用干净 venv 从 wheel 安装。
3. 在包目录之外验证已安装 CLI 命令的运行行为。

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实验中的脚本文件:

1. `Scripts/q09/src/greetlab/__init__.py`
2. `Scripts/q09/src/greetlab/cli.py`
3. `Scripts/q09/pyproject.toml`